



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01/08/2026

In addition to Part-I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No. : MATH F312
Course Title : Ordinary Differential Equations
Instructor-In-Charge : BALRAM DUBEY, Email: bdubey@pilani.bits-pilani.ac.in, Chamber No: 6067-B
Instructors :

1. COURSE DESCRIPTION:

Existence and uniqueness theorems; properties of linear systems; behavior of solutions of nth order equations; asymptotic behavior of linear systems; stability of linear and weakly nonlinear systems; conditions for boundedness and the number of zeros of the nontrivial solutions of second order equations; stability by Lyapunov's direct method; autonomous and non-autonomous systems.

2. SCOPE & OBJECTIVE:

Ordinary Differential Equations frequently occurs as mathematical models in many branches of science, engineering and economy. For a mathematician confronted with such a model there are a number of issues to address and various approaches to choose from:

Is the problem well-posed? Do you expect the differential equation to have a solution? If so, is there a unique solution satisfying the given initial or boundary conditions?

Can you find an explicit, analytical solution? This is only possible in rare circumstances.

Geometric or qualitative methods: These methods give insights into general, qualitative features of solutions and do not require solving the differential equation.

Stability and dependence on parameters: Having obtained a solution by any method, we would like to know how the solution changes if we change the initial data by a small amount (stability analysis) and if we change parameters in the differential equation (parameter dependence). Course helps deeper understanding of the complicated models that are there in the real life.

3. Course learning outcomes:

- CLO1. Existence and uniqueness of ODEs of nth order
- CLO2. Linear and nonlinear behavior of system of ODEs
- CLO3. Applications of ODEs in real-world problems



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4. TEXT BOOK:

S Ahmad & M R M Rao : Theory of Ordinary Differential Equations with Applications in Biology and Engineering, East West Press, 1999.

REFERENCE BOOKS:

R1. The qualitative theory of ordinary differential equations, an introduction, Fred Brauner and John A Nohel, Dover Publications, 1969.

R2. Theory of Ordinary Differential Equations, E.A. Coddington & N. Levinson, Tata McGraw- Hill, 1972.

5. Course Format & Weekly Workflow

Wk no.	Module Number	Description	Ref.	Learning Outcome
1-3	M1.Existence, Uniqueness and continuation of solutions	L1-L2. Overview of the course, Some important definitions	1.1-1.2	(i) Able to visualize existence and uniqueness of solutions of scalar and vector differential equations.
		L3-L9 Solutions of Scalar Differential equations, Existence Theorems, Differential & Integral Inequalities	1.3-1.5	(ii) Able to use applications of Gronwall-Reid- Bellman inequality.
4-5	M2. Linear Systems	L10-L13 Introduction to Linear Systems, FSS, Properties of Linear Homogeneous Systems, Inhomogeneous Linear Systems	2.1-2.3	(i) Able to know the properties of system of linear differential equations. (ii) Able to analyze the behavior of solutions of system of linear differential equations (without solving the equations).





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		L14-L15 Behavior of Solutions of nth order Linear Homogeneous Equations, Asymptotic Behavior	2.4-2.5	
6-8	M3. Stability of linear and weakly nonlinear systems	L16-L19 Introduction to stability, Continuous dependence and stability properties of Solutions	3.1-3.2	Able to know the stability behavior of linear and weakly nonlinear systems using eigenvalue theory.
		L20-L24 Linear Systems, Weakly non linear Systems, Two Dimensional Systems	3.3-3.5	
9-12	M4. Stability by Liapunov's Second Method	L25-L34 Introduction to stability by Liapunov's Direct Method, Autonomous systems, Non Autonomous Systems	5.1-5.3	Able to know the stability behavior of linear and nonlinear system of differential equations by Liapunov's Direct Method (without
13-14	M5. Properties of second order differential equations	L35-L40 Second order differential equations: boundedness of solutions, Oscillatory equations, Applications to some Classical equations	4.1-4.5	Able to know the boundedness and oscillatory properties of 2 nd order differential equations without solving them.





6. Evaluation Scheme

Component	Duration	Weightage (%)	Date & Time	Nature
Midsem Test	90 mins	35%	As announced in Time Table	OB
Quiz*	20 mins each	18%	Announced/unannounced	CB/OB
Assignments	-	2%		
Comprehensive Examination	3 hours	45%	As announced in Time Table	CB

* 3 quizzes will be conducted and best 2 will be considered.

7. MAKE-UP Policy: Prior permission is needed for makeup. Makeup will be given only in a very genuine case. **No makeup will be granted in Quizzes/Assignment component.**

8. Criterion for NC: 10% of the total marks or 20% of the average, whichever is Lower.

9. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

10. NOTICES: All notices will be displayed on the online Nalanda NB. Normally information will also be conveyed in the class.

Instructor-in-charge
MATH F312

